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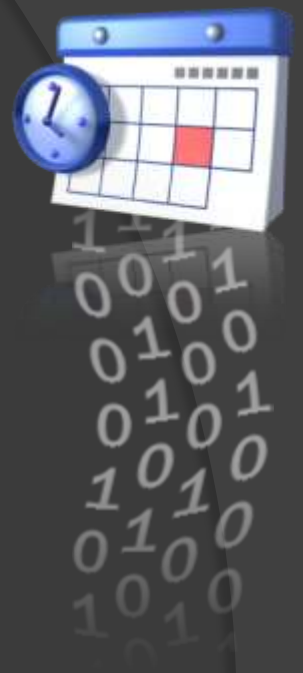
MICROSOFT'S RIA PLATFORM AND SILVERLIGHT 2 IN DETAIL

Agenda



- Microsoft RIA Technologies
- Silverlight 2 in Detail
  - Introduction
  - Architecture
  - Technical Details

Microsoft RIA Technologies

REACH INTERNET APPLICATION*

ASP.NET AJAX

RICH INTERNET APPLICATION

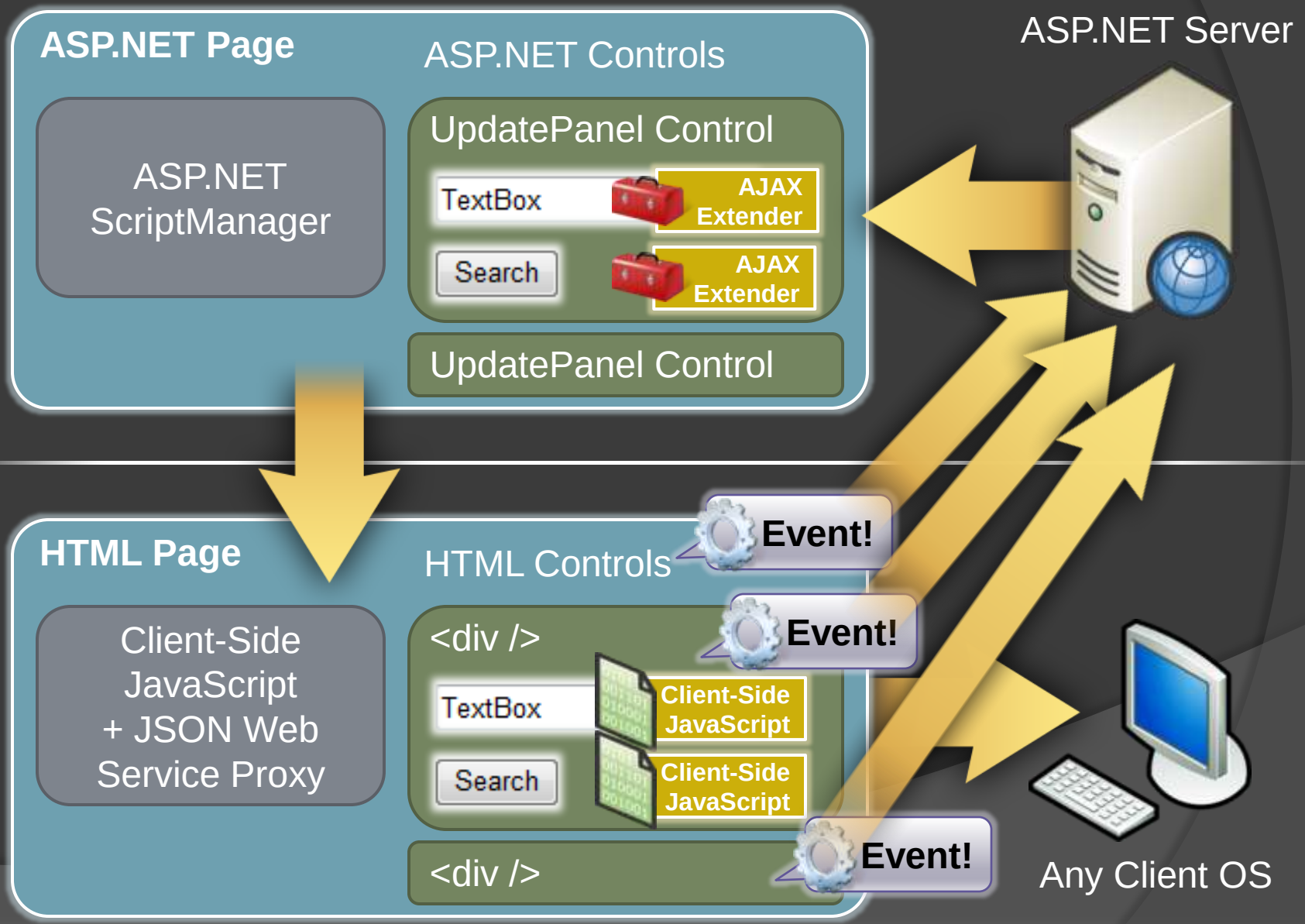
Silverlight 2

"ULTRA-RICH" INTERNET APPLICATION*

Windows Presentation Foundation (IE Hostable)

** No official term*

ASP.NET AJAX Framework



Forrester Research Study

FOR VENDOR STRATEGY PROFESSIONALS Length: 8 pages

March 20, 2008

Ajax Disappoints Power Users Looking For Web 2.0-Style Business Apps

Some Words Of Caution For Business Application Vendors Moving To Ajax

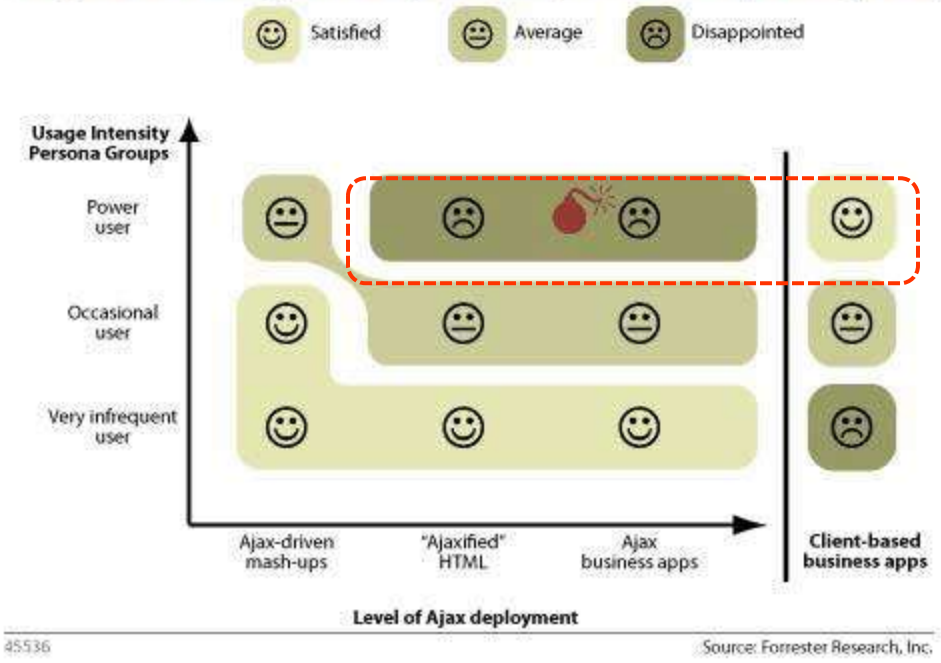
by Stefan Ried, Ph.D.

with Thomas Mendel, Ph.D., Jeffrey S. Hammond, Reedwan Iqbal

EXECUTIVE SUMMARY THIS IS A DOCUMENT EXCERPT

A great user experience is one of the most compelling and important characteristics of a modern business application. Thus, application vendors are under permanent pressure to offer new user interface (UI) technologies and introduce rich Internet applications (RIAs) instead of traditional Windows clients. Strategy professionals at application vendors should become familiar with the concept of persona usage intensity, which characterizes user expectations and applies them to the widely used Ajax technology. In contrast to occasional users, most power users are disappointed by Ajax-based RIAs, and their technical limitations are hard to overcome. Adobe and Microsoft have recently released new client technologies for production, and application vendors should evaluate these now.

Figure 1: Matching The Usage Intensity Persona Groups To Ajax Deployments



http://www.forrester.com/Research/Document/Excerpt/0,7211,45536,00.html

Silverlight 2

INTRODUCTION

Microsoft Silverlight

cross-browser
cross-platform
cross-device

Plug-In

for building and delivering the next generation of

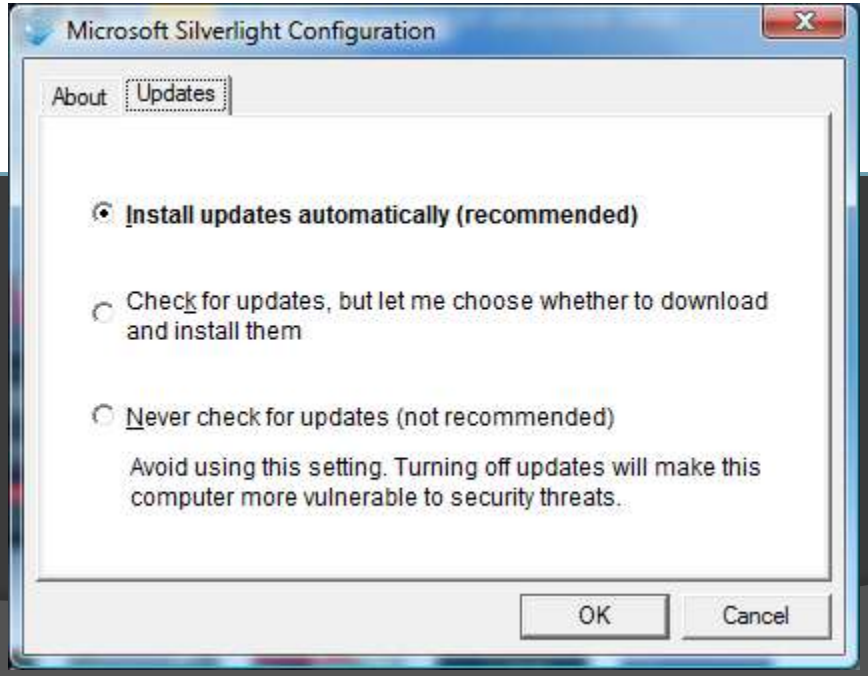
.NET based

Media Experiences
and
Rich Interactive Applications

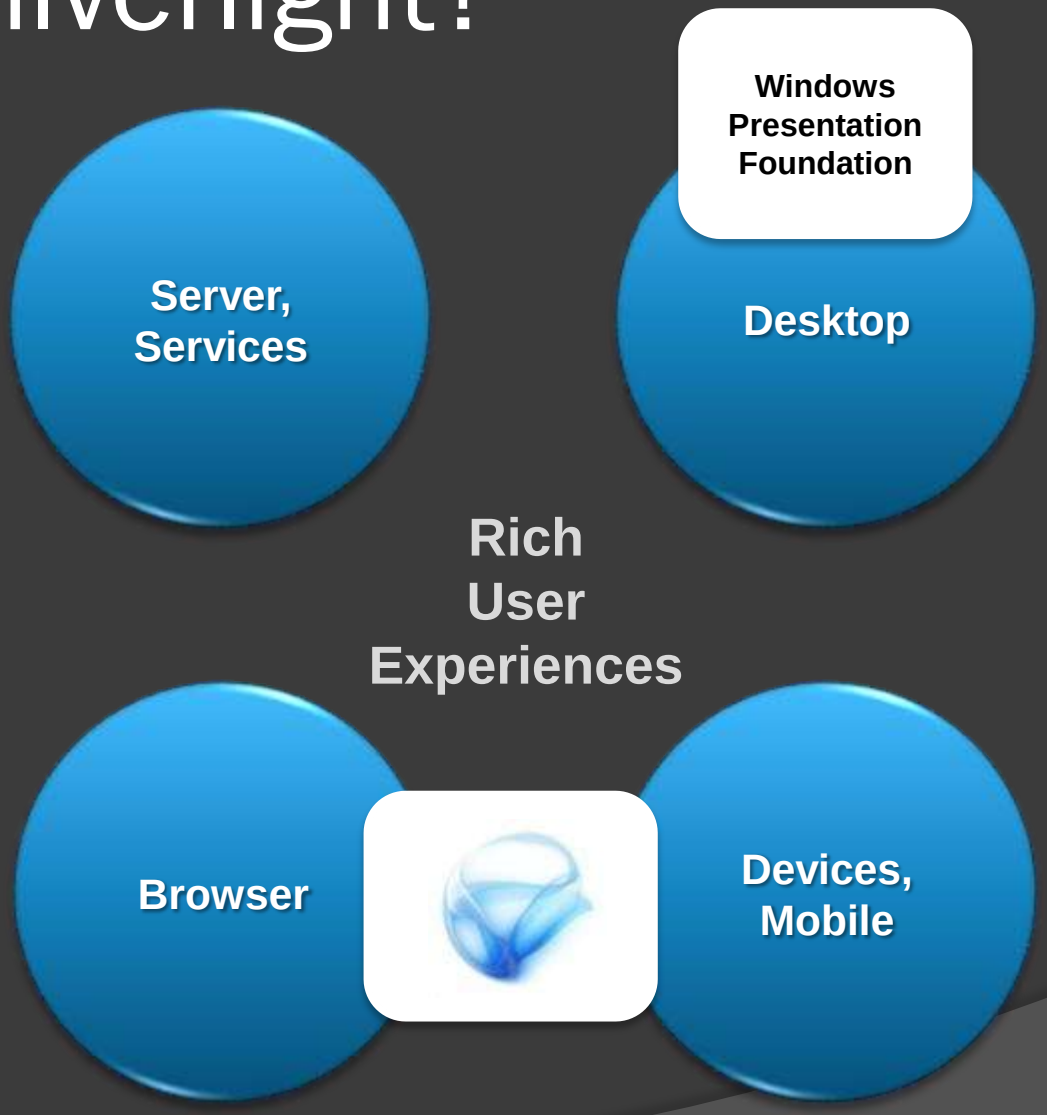
for the Web

Microsoft®
Silverlight™

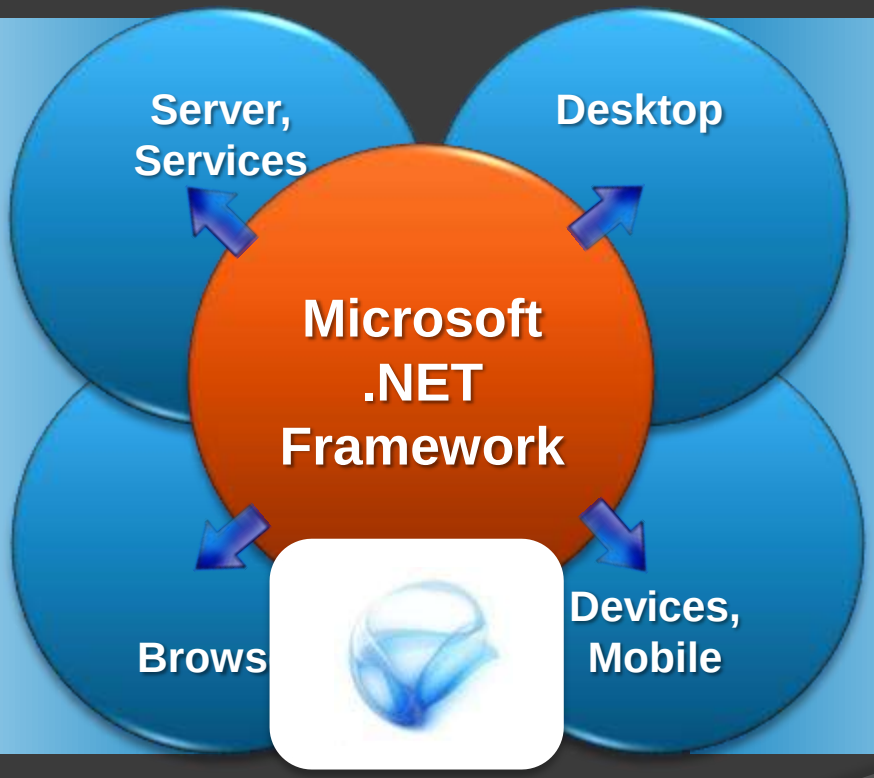
Silverlight Installation and Update



Why Silverlight?



Silverlight is a .NET Technology

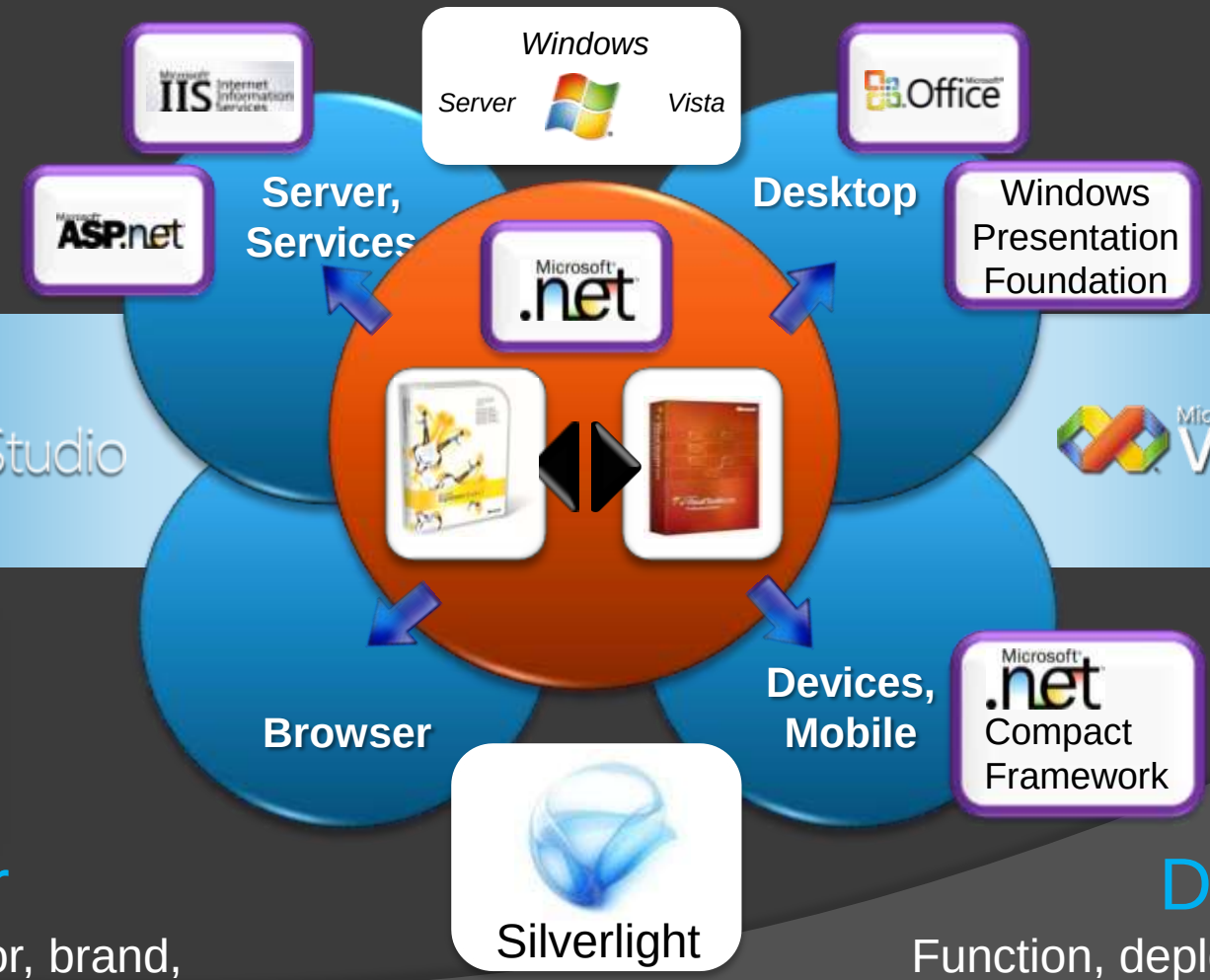


.NET Extends
Developer's Reach

Designer & Developer Tooling

Designer

Look, behavior, brand, and emotional connection



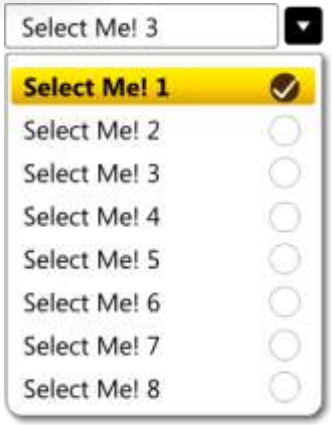
Developer

Function, deployment, data, security, operational integrity

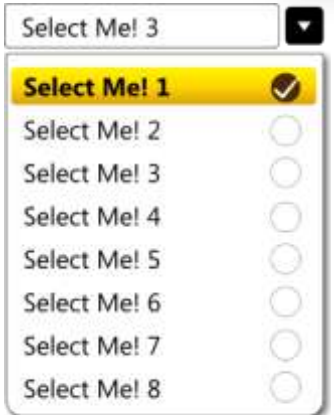
Unifying Design / Development

Designer
Look, behavior, brand,
and emotional connection

Developer
Function, deployment, data,
security, operational integrity



XAML



XHTML
CSS / XSLT
XML
ASP.NET
Javascript
AJAX

Silverlight 2

ARCHITECTURE

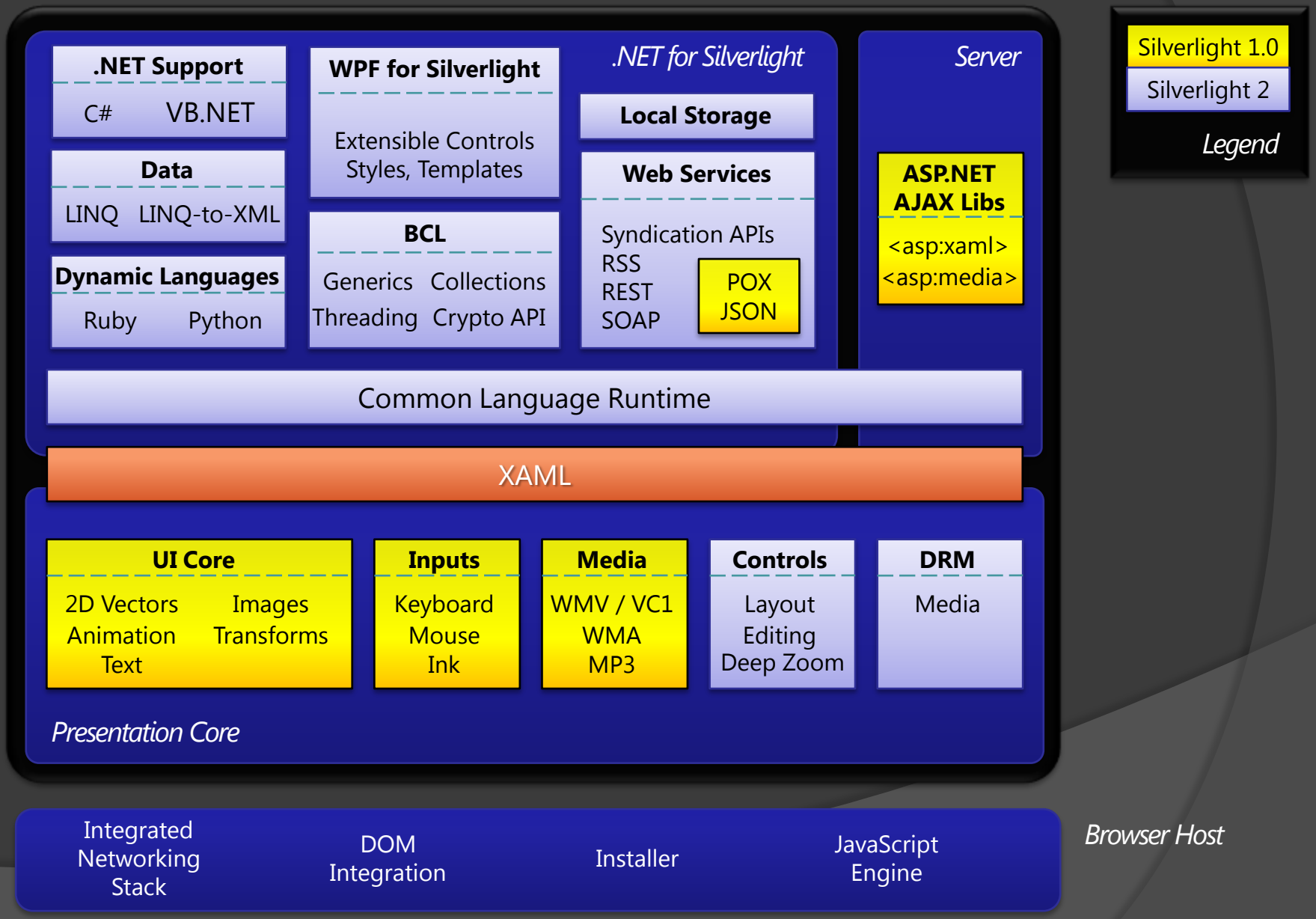
Silverlight Versions Compared

Silverlight 1

Silverlight 2

WPF

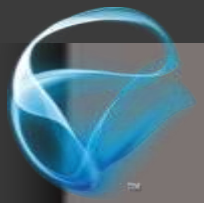
Silverlight 2 Architecture Map



Silverlight 2

TECHNICAL DETAILS

Data Driven Apps



Microsoft Silverlight

Asynchronous Calls Only

1. **Build URL**
2. **Make Request with WebClient / HttpWebRequest**
3. **Work with Request/Response Data (XML or JSON)**

"Mashup APIs"
"Web APIs"
REST Services
"POX" Services
JSON Services

Human-Readable Service Description

1. **Use built-in SyndicationFeed Class**
2. **Access Items through SyndicationItem Collection**
3. **Can Read / Write Feeds**

RSS / ATOM Feeds

Standards-Based Service / API

```
WeatherServiceProxy p = new WeatherServiceProxy();
p.
```

- GetStormWarnings
- GetWeather
- GetWeatherForecast

1. Add Service Reference
2. Works like a local Class

WCF Service
SOAP Service

Self-Describing Service (WSDL)

Local Isolated Storage

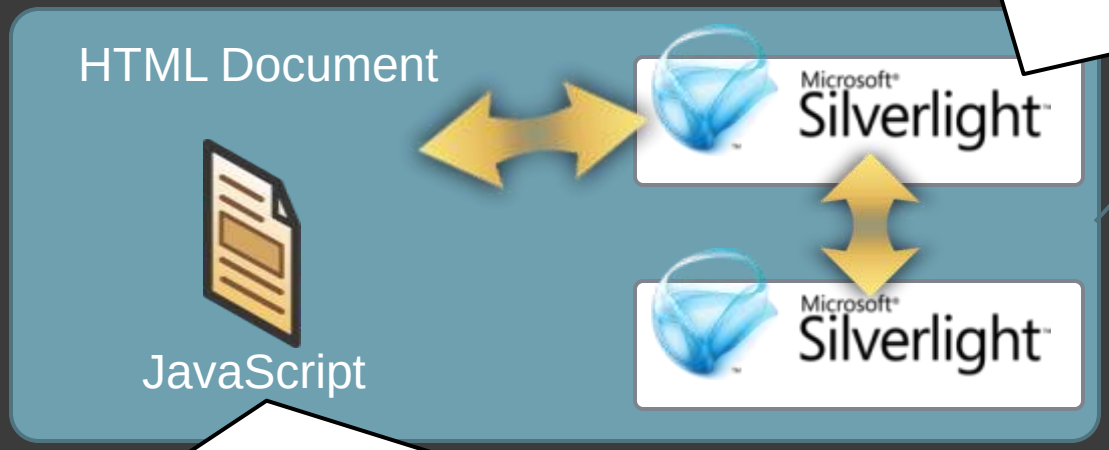
- “Isolated Storage”
- Like Cookies but more Powerful
- Not Expiring, Provide full Control
- Provides initial 100KB of Application specific Local Storage
- Can be expanded per app by prompting the user, up to “unlimited”.
- Exposed through .NET IsolatedStorage APIs

DEMO

Data Driven Silverlight Application

Full HTML DOM Integration

```
if (HtmlPage.Document.DocumentUri.AbsoluteUri.EndsWith("TestPage.html") != true)
  return;
HtmlElement btnUC = HtmlPage.Document.GetElementById("buttonUCtext");
btnUC.AttachEvent("onclick", new EventHandler<HtmlEventArgs>(this.OnUCtextClicked));
btnUC.SetProperty("disabled", false);
btnUC.SetAttribute("value", "This text from the Silverlight Side");
```



Write Business Logic in Managed Code and call from JavaScript
→ Very easy to debug

```
<script type="text/javascript">
  function onClickButton()
  {
    var control = document.getElementById("Xaml1");
    control.content.MyControl.Hello("Silverlight");
    control.content.findName("tb").Text = "Hello Silverlight";
  }
</script>
```

Silverlight Web Media Ecosystem

Create

Microsoft Expression Studio

Expression Blend for creating media experiences

Expression Media Encoder for publishing with Silverlight

+ Existing ecosystem of WMV technology partners / solution providers

Distribute

Windows Server 2008

Up to 2x streaming scalability over Windows Server 2003

New IIS7 Media Pack with bit-rate throttling for lower cost for downloaded media

New Secure content delivery over SSL and Cache/Proxy support

Experience

Microsoft Silverlight

Interactivity, video, and animation in browser and/or full screen

Seamless, Fast Installation for End Users

Consistent experience on Mac and Windows

Silverlight Media Features

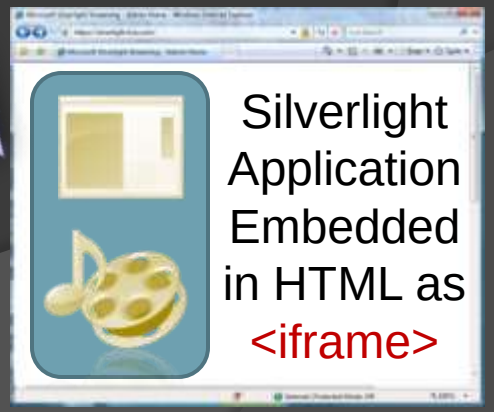
- Self-contained media playback
- Non-rectangular, semi-transparent video with overlays
- Alpha video
- Customizable transport controls
- Media markers / script commands
- ASX playlist support
- Closed captioning support with Expression Encoder
- Progressive download and Windows Media Services support
  - Live and on demand streaming
  - Bandwidth throttling with IIS 7.0
- Content protection

Silverlight Streaming

Silverlight Application

Silverlight Video Asset (VC-1)

http://streaming.live.com



Microsoft Visual Studio 2008

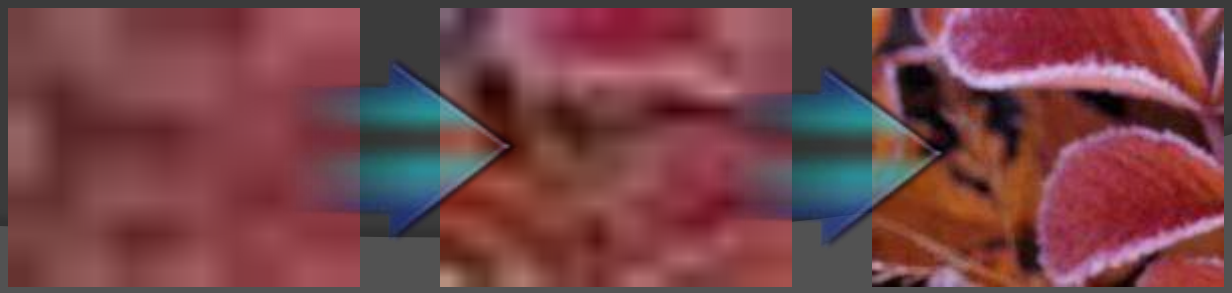
Microsoft Expression

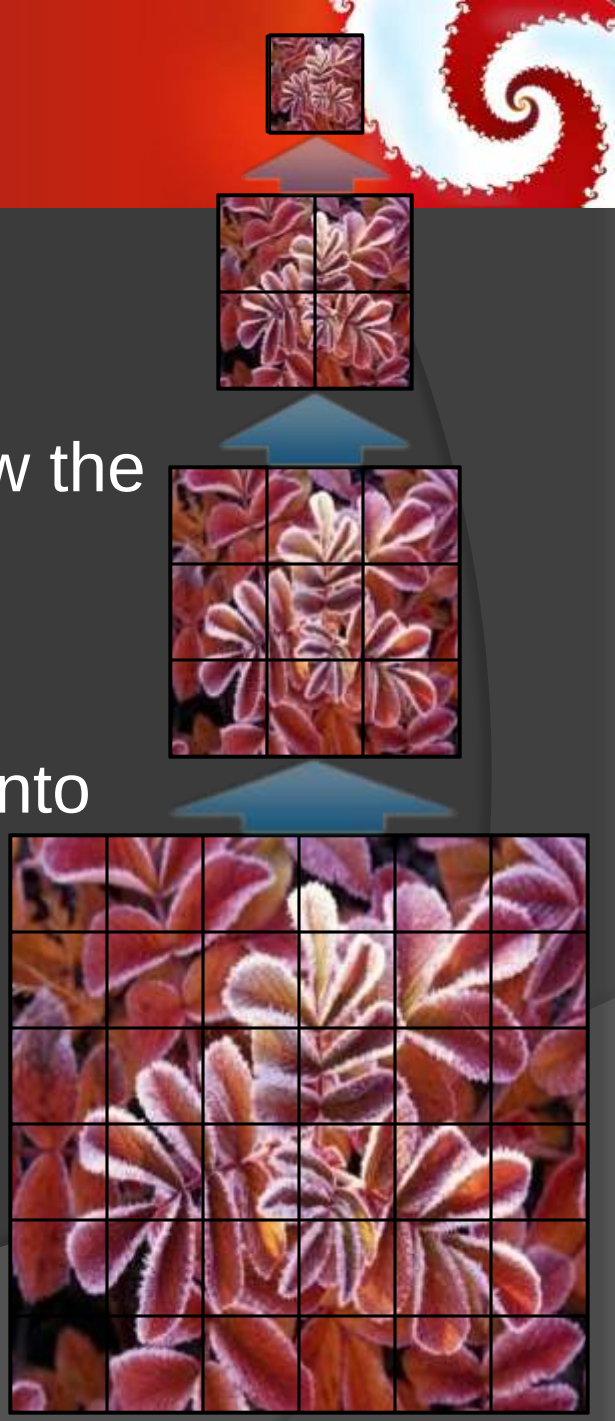
Blend

Encoder

Deep Zoom

- Seamless Viewing & Zooming of huge Images
- Loads only the Data necessary to show the Part of an Image the User is Viewing
  - Lowest Resolution Tiles shown first
  - Download Higher quality Tiles and blend in
- Deep Zoom Composer breaks Image into Tiles
  - Export as Deep Zoom Image or Collection
  - Exposed through MultiScaleImage Control





DEMO

Deep Zoom

The Cost of Silverlight

Development

Full SDLC:
Visual Studio Team System

Developer: $5'000
Suite: $10'000

VB/C#/XAML:
Visual Studio 2008

Std: $300
Pro: $800

XAML Design:
Expression Blend

$500

Any Text Editor

Free non MS

Hosting

Advanced Streaming:
Windows Server 2008
+ Windows Media Services

Enterprise: $5000
Standard: $1000
Web: $500

Free

Any Web Server

Free non MS

Running

Plug-In

Free

Roadmap

Mix08 (March 08)	Summer 08	Beyond…
Silverlight 2 Beta 1	Silverlight 2	
Silverlight Mobile Demoed on Symbian & Windows Mobile	Silverlight Mobile 1 CTP (no .NET)	Silverlight Mobile 2 CY09 (with .NET)
Expression Studio 2 Beta Expression Blend 2.5 Preview	Expression Studio 2	Expression Blend 2.5
Silverlight 2 Tool Alpha for Visual Studio 2008 Beta		Silverlight 2 Tools for Visual Studio 2008

Silverlight 2: Links

- Main Silverlight Site
  - http://silverlight.net/default.aspx
- Silverlight Developer Center on MSDN
  - http://msdn.microsoft.com/silverlight/
- “Get Started” with Silverlight – All needed Downloads
  - http://silverlight.net/GetStarted/
- Silverlight Learning Center
  - http://silverlight.net/Learn/
- Silverlight Showcases
  - http://silverlight.net/Showcase/

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